CARD 1 OF 8 TWO ZONES **QUESTION**

NFPA 13 splits obstructions under a sprinkler into **two zones**. Where is the dividing line, and what goes wrong in each zone?

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CARD 2 OF 8 THREE TIMES RULE **QUESTION**

A standard upright sits near a 6 in. wide pipe rack in Zone 1. How far away does the **Three Times Rule** want the head, and where does the rule stop?

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CARD 3 OF 8 LIGHT & ORDINARY HAZARD **QUESTION**

In a **light or ordinary hazard** office, does the Three Times Rule apply to a light fixture, a duct and a 2 in. sprinkler main?

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CARD 4 OF 8 HEADS ON BOTH SIDES **QUESTION**

A beam or duct up to **4 ft wide** splits your layout. Instead of moving heads off it, what layout does NFPA 13 allow?

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*FSM · Sprinkler Fact Cards — NFPA 13 (2019) Obstructions: The Two Zones · sheet 1 FRONT — print double-sided, **flip on LONG edge** · cut on the dashed lines*

CARD 2 OF 8 THREE TIMES RULE **ANSWER**

At least 3 × the obstruction's biggest dimension. 6 in. wide → 18 in. away.

Cap: you never need more than **24 in.** of clearance.

Columns: that 24 in. cap does **not** apply to vertical obstructions.

NFPA 13 (2019) · Ch. 10

Trap: using it on a solid beam. The Annex says the rule fails there because water can't get over and under it. Beams go to the Beam Rule.

CARD 1 OF 8 TWO ZONES **ANSWER**

The line is 18 in. below the deflector.

Zone 1 (18 in. or less): the obstruction keeps the spray pattern from **forming**.

Zone 2 (more than 18 in.): the pattern has formed, but the obstruction keeps the water from **reaching the hazard** below it.

Either way, move the head or add heads.

NFPA 13 (2019) · Ch. 9

Trap: measuring from the ceiling. Both zones are measured down from the **deflector**.

CARD 4 OF 8 HEADS ON BOTH SIDES **ANSWER**

Put a head on each side of it.

Condition: from the obstruction's **centerline** to each head can be no more than **half** the spacing you're allowed between heads.

Example: 15 ft allowed spacing → each head within 7½ ft of the centerline.

NFPA 13 (2019) · Ch. 10

Trap: measuring to the edge of the obstruction. It's the **centerline**.

CARD 3 OF 8 LIGHT & ORDINARY HAZARD **ANSWER**

No, no and no.

LH and OH: only **structural members** count for the Three Times Rule. A light or a duct isn't structure.

Sprinkler pipe under 3 in. is exempt from it in any hazard.

Annex: the **Beam Rule** and the **Wide Obstruction Rule** still apply to those same items.

NFPA 13 (2019) · Ch. 10 · Annex A

Trap: "nonstructural, so no obstruction rules." Only the Three Times Rule drops out.

sheet 1 BACK — columns are intentionally mirrored for long-edge duplex · card numbers must match front to back

CARD 5 OF 8 WIDE OBSTRUCTION RULE **QUESTION**

Zone 2: a duct hangs 3 ft below the deflectors. When does it need **sprinklers underneath**?

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CARD 6 OF 8 HEADS UNDER IT **QUESTION**

You're adding heads under a 6 ft wide duct. Where do they go, and **what kind of head**?

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CARD 7 OF 8 OPEN GRATING **QUESTION**

A mezzanine has **open grate flooring** 8 ft wide. The water can fall through it, so does it still need sprinklers below?

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CARD 8 OF 8 EXCEPTIONS & STACKS **QUESTION**

Two wide-obstruction cases: a **noncombustible** 6 ft wide platform sitting low to the floor, and three 5 ft wide ducts **stacked** on top of each other. What does each need?

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CARD 6 OF 8 HEADS UNDER IT **ANSWER**

Deflector within 12 in. of the duct's bottom.

Location: under it, or alongside it no more than **3 in.** from its outside edge. Heads alongside must be **intermediate level / rack type**.

Type: same as the ceiling heads (spray under spray, ESFR under ESFR). Spray heads are allowed under overhead doors.

NFPA 13 (2019) · Ch. 9 · Ch. 10

Trap: a plain pendent tucked beside the duct. Beside it, NFPA 13 calls for an intermediate level / rack-type head.

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CARD 5 OF 8 WIDE OBSTRUCTION RULE **ANSWER**

When it's fixed in place and over 4 ft wide.

Annex: at more than 18 in. down the spray has formed, so anything **up to and including 4 ft** wide needs nothing below it.

Not fixed? Things that move, like a conference table, don't need heads under them.

NFPA 13 (2019) · Ch. 9 · Annex A

On the job: a 5 ft wide duct, a catwalk, a deck or a cutting table in a shop all trip it. A 4 ft duct exactly does not.

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CARD 8 OF 8 EXCEPTIONS & STACKS **ANSWER**

Low noncombustible platform: no heads under it, if it's noncombustible and its bottom is **24 in. or less** above the floor or deck.

Stacked ducts (Annex): one level of heads under the bottom duct may not be enough. Extra levels of protection can be needed.

NFPA 13 (2019) · Ch. 9 · Annex A

Trap: the exception needs **both** conditions. A combustible wood platform low to the floor gets no pass.

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CARD 7 OF 8 OPEN GRATING **ANSWER**

Yes. Open grate floors over 4 ft wide need protection underneath.

The heads under it must be **intermediate level / rack type** or otherwise **shielded**, so water from above doesn't cool them.

Annex: slatting a deck or swapping in open grating is **not** a substitute for sprinklers below.

NFPA 13 (2019) · Ch. 9 · Annex A

Trap: "it's see-through, the ceiling heads will get it." Over 4 ft wide, they won't be enough.

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